

**TransitTrack: A Comprehensive Web-Mobile Application for Streamlining Public
Transportation Operations in Cebu City.**

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By

Renzo Miguel M. Deiparine

Kyle C. Bonghanoy

Christian Javidil A. Sy

Adviser

Mr. Temothy Homecillo

CHAPTER 1

ABSTRACT

The growing inefficiency and congestion in the Philippines' public transportation system, particularly in Cebu City's South and Northbound terminals, have significantly impacted the commuting experience. Despite attempts to modernize through emerging technologies, current systems still face challenges in managing road and terminal traffic, resulting in delays and disorganized operations. This capstone project presents TransiTrack, a web-mobile application designed to enhance the coordination and efficiency of public transit services. By offering tailored features for commuters, drivers, terminal managers, and bus operators, TransiTrack aims to streamline transit flow, improve real-time communication, and promote a safer, more organized transportation environment. The system's integration with existing operations seeks to provide a smarter and more responsive solution to address long-standing issues in the urban transit landscape.

INTRODUCTION

The rapid urbanization and economic growth in Cebu City, Philippines, have exacerbated the inefficiency and congestion problems in its public transportation system, particularly at the South and Northbound bus terminals. Despite various modernization efforts, including the introduction of the Bus Rapid Transit (BRT) system, the city continues to face significant challenges in managing road and terminal traffic, leading to delays, disorganized operations, and commuter dissatisfaction (Alano, 2023). The persistent traffic congestion is compounded by narrow roads, increasing vehicle numbers, and limited infrastructure expansion options, making traditional solutions like road widening and flyover construction both costly and insufficient (Cebu Daily News, n.d). Recognizing that public transportation is key to addressing these issues, there remains a critical need for innovative, integrated solutions that enhance coordination among commuters, drivers, terminal managers, and bus operators while improving real-time communication and transit flow.

This capstone introduces TransiTrack, a web-mobile application designed to address these challenges by streamlining public transit operations and improving the overall commuting experience in Cebu City. By integrating with existing systems and offering tailored features for all stakeholders, TransiTrack aims to create a smarter, more responsive transportation environment that mitigates congestion, enhances safety, and promotes more organized terminal management. This study explores the potential of TransiTrack to contribute to the sustainable development of Cebu's urban transit landscape, complementing ongoing infrastructure projects and supporting the city's

vision for a more efficient and commuter-friendly public transportation system (South Pole Central Hotel, 2025).

RATIONALE OF THE STUDY

The public transportation system in Cebu City, particularly at the South and Northbound terminals, is increasingly burdened by inefficiency and congestion, which significantly affect the daily commuting experience of thousands of passengers. Despite various modernization efforts, challenges such as traffic delays, disorganized terminal operations, and poor communication among commuters, drivers, and terminal managers persist. These issues not only cause inconvenience but also contribute to safety risks and reduced productivity. Research shows that urban transportation inefficiencies are a common challenge in rapidly growing cities, where sustainable mobility solutions are urgently needed to address environmental and social impacts (Cervero, 2013).

Mobile and web-based applications have demonstrated effectiveness in improving public transit systems by providing real-time information, optimizing route management, and enhancing communication among stakeholders. For example, studies have found that real-time transit information can increase ridership and reduce wait times by enabling better trip planning (Zhou & Levinson, 2017). Furthermore, the integration of innovative mobility technologies such as ride-hailing and carsharing within

urban transport ecosystems has been shown to improve operational efficiency and commuter satisfaction (Shaheen & Cohen, 2019).

This project proposes TransiTrack, a web-mobile application designed to address the challenges faced by Cebu City's public transportation terminals. By offering tailored features for commuters, drivers, terminal managers, and bus operators, TransiTrack aims to streamline transit flow, improve real-time communication, and promote a safer, more organized transportation environment. Leveraging emerging technologies aligns with global best practices and offers a promising pathway to create a smarter, more responsive urban transit system that supports sustainable mobility and enhances the quality of life for commuters.

REVIEW OF RELATED LITERATURE

Recent developments in Cebu City's public transportation terminals highlight the persistent management and operational issues that directly affect service quality and commuter experience. As of July 2025, no official terminal managers have been appointed for the Cebu South Bus Terminal (CSBT) and Cebu North Bus Terminal (CNBT). Instead, volunteers and temporary staff are handling daily operations, leading to concerns about accountability, consistency, and efficiency in terminal management (Villa, 2025). This situation underscores the importance of structured management systems and the risks posed by informal or ad hoc arrangements.

The absence of official managers has also reignited debates over the efficiency of current ticketing systems. Complaints from both bus operators and passengers about long queues and system inefficiencies have prompted considerations to revert from digital to manual ticketing processes. This reflects broader challenges in integrating technology into public transport operations, where poorly implemented systems can exacerbate, rather than resolve, existing inefficiencies (Villa, 2025). Literature on urban transit modernization stresses the need for robust digital solutions that are user-friendly and responsive to local operational contexts (Zhou & Levinson, 2017).

Urban centers in the Philippines, such as Cebu City, face severe traffic congestion and outdated infrastructure that limit public transport efficiency. According to Del Castillo and Dela Cruz (2020), the lack of sufficient public transport routes and aging fleets contribute to long commute times and unreliable services, negatively impacting commuters' daily lives. Traffic congestion imposes significant economic costs and environmental harm. A study by the Asian Development Bank (ADB, 2019) estimated that traffic congestion in Metro Manila alone costs the economy billions of pesos annually and increases greenhouse gas emissions, which exacerbate public health issues. Real-time transit information systems have been shown to reduce wait times and improve passenger satisfaction by providing accurate schedules and updates. Ferris, Watkins, and Borning (2010) found that such systems empower commuters to plan trips more effectively, reducing uncertainty and perceived waiting time.

Ziółko and Kołodziej (2024) conducted a comprehensive study analyzing mobile applications supporting public transport in smart cities using Poland as a case study. Their research, based on a survey of 1,000 respondents, revealed that the majority of users rely on mobile applications for public transportation, with the most important functionalities being route determination from point A to point B, timetable availability, and total travel time calculation. The study identified key areas for improvement, including expanding app availability to more cities and enhancing ticket purchasing capabilities.

SIGNIFICANCE OF THE STUDY

The persistent challenges in Cebu's public transportation terminal including the absence of formal management systems, inefficient ticketing processes, and inadequate facility maintenance underscore the necessity for integrated digital solutions. These operational gaps not only hinder commuter experience but also impede the overall effectiveness and safety of transit services. Addressing these issues is critical for the sustainable development of urban mobility in Cebu City.

TransiTrack, as a web-mobile application, offers a comprehensive approach to resolving these challenges by introducing real-time communication channels, structured management interfaces, and streamlined ticketing mechanisms. The implementation of such technology-driven platforms is expected to enhance operational transparency, accountability, and efficiency within terminal operations. Furthermore, by facilitating better coordination among commuters, drivers, terminal managers, and bus operators,

TransiTrack aims to improve commuter satisfaction and promote safer, more organized transit environments.

This study is significant as it contributes to the growing body of literature advocating for the adoption of digital management tools in public transportation systems. By demonstrating the practical benefits of TransiTrack, the research supports the case for technology integration as a means to address longstanding inefficiencies and to foster a more responsive and commuter-friendly urban transit system (Shaheen & Cohen, 2019).

REVIEW OF RELATED WORKS

Case Studies in the Philippines:

The Cebu Bus Rapid Transit (BRT) project exemplifies ongoing efforts to modernize urban transit in the Philippines. The Department of Transportation (2023) reported that the BRT aims to decongest roads, provide safer mobility, and serve up to 160,000 passengers daily, reflecting a broader push toward digital and infrastructure upgrades.

Digital Ticketing and Fare Collection:

The adoption of automated fare collection and digital payment systems has streamlined boarding processes, reduced queues, and minimized cash handling. Shaheen and

Cohen (2019) noted that these systems support operational efficiency and sustainability, as well as enhance the overall user experience.

Optimizing Transportation with Smart Bus Management Systems

Recent research by Shreya et al. (2024) focused on integrating RFID technology, GPS tracking, and IoT sensors for comprehensive bus management. Their system includes touch sensors for seat availability detection, RFID-based ticketing, and real-time data transmission to cloud platforms via Google Sheets API. The study demonstrated how modern sensor technologies can enhance passenger experience while improving operational efficiency.

PROJECT OBJECTIVES

General Goals :

The general goal of this capstone project is to develop and implement TransiTrack, a comprehensive web-mobile application designed to enhance the coordination, efficiency, and user experience of public transportation services at Cebu City's South and Northbound bus terminals. The project aims to address persistent issues of congestion, management inefficiency, and poor communication among commuters by leveraging digital solutions that streamline operations and promote a safer, more organized transit environment.

Specific Goals :

- Provide commuters with real-time bus tracking, route planning, digital ticketing, and facility feedback functionalities, improving convenience and reducing wait times.
- Equip drivers with up-to-date schedule information, route assignments, and direct communication tools for more efficient operations.
- Enable terminal managers to utilize digital dashboards for incident reporting, performance monitoring, and overall terminal management.
- Offer bus operators access to operational analytics, fleet management tools, and compliance tracking to support data-driven decision-making.
- Integrate the application with existing terminal operations and modernization initiatives, ensuring compatibility and seamless adoption.
- Assess the effectiveness of TransiTrack in improving terminal efficiency, commuter satisfaction, and stakeholder coordination through pilot implementation and user feedback.

Non-Functional Goals:

Reliability: The system must provide accurate, up-to-date information and maintain high availability, minimizing downtime and service interruptions.

Security: Implement robust security measures to protect user data, ensure privacy, prevent unauthorized access or misuse of information, emergency contacts and incident reporting

Usability: Ensure the application features an intuitive, user-friendly interface accessible to users with varying levels of digital literacy.

SCOPE AND LIMITATIONS OF THE STUDY

The study focuses on developing and implementing TransiTrack, a comprehensive web-mobile application designed to improve public transportation services at Cebu City's South and Northbound bus terminals. The study aims to enhance coordination, efficiency, and user experience by addressing key issues such as terminal congestion, management inefficiency, and poor communication among commuters, drivers, and terminal managers. It includes enabling terminal managers to use digital dashboards for incident reporting, performance monitoring, and overall terminal management. The study also explores digital ticketing and fare collection systems to streamline boarding processes and reduce queues.

The limitations of the study likely include the focus on only two specific bus terminals in Cebu City, which may restrict the generalizability of the findings to other locations. The study may also be limited by the current infrastructure and technology adoption level within the terminals and the users' willingness to adapt to new digital systems. Challenges related to real-time data accuracy and comprehensive integration with existing transportation networks might also constrain the system's effectiveness. External factors such as traffic conditions and policy changes in the transportation sector could influence the results and operational success of the application.

Overall, the study is scoped to deliver a practical digital solution targeting specific operational challenges at key Cebu City terminals while acknowledging constraints in geographic coverage, technological readiness, and external environmental factors.

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